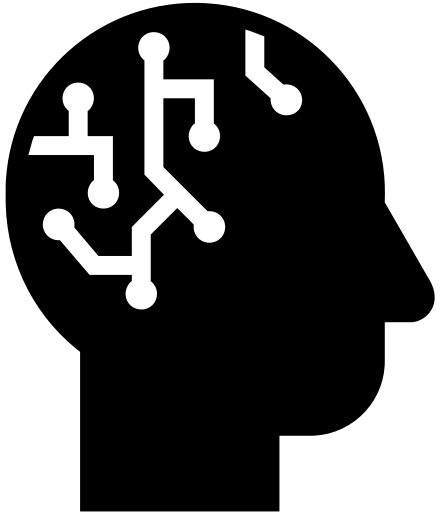
Karl Estampador, Charlotte Lai, Merry Liu

The Impact of Region of Origin on Income: Analyzing Earnings Disparities Among Canadian Immigrants



This project aims to investigate the extent to which income differences exist among immigrants from different origins and to evaluate whether these disparities persist even after accounting for various factors.

This project can provide a comprehensive understanding of the association between regional origin and income level, revealing whether immigrants from certain regions face comparative disadvantages or enjoy advantages in terms of earnings.

Our key research question seeks to understand whether immigrants from certain regions tend to earn higher incomes than others after settling in Canada.

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Methodology & Data description

- Approximately 980,000 samples from the 2021 Census Public Use Microdata File Individuals File from Statistics Canada.
- We use total income (TOTINC) as the dependent variable and place of birth (POB) as the main independent variable. Other control variables are also included for the purpose of reducing omitted variable bias and isolating the effect of POB on TOTINC.
- We conducted VIF test for multicollinearity for robustness check and the Table 4 shows the result.
- We use Log-Linear model as below

Table 4—: Generalized Variance Inflation Factors (GVIF) for Key Predictors

Variable	GVIF	Df	VIF score	Other Predictors
POB	6.091e+01	31	1.068527	HDGREE, CIP2021, LOCSTUD, noc21, KOL, AGEIMM, Gender, MarStH, HHSIZE
HDGREE	-2.027e+17	12	NaN	POB, CIP2021, LOCSTUD, noc21, KOL, AGEIMM, Gender, MarStH, HHSIZE
CIP2021	1.340e+17	12	5.171636	POB, HDGREE, LOCSTUD, noc21, KOL, AGEIMM, Gender, MarStH, HHSIZE
LOCSTUD	1.962e+17	7	17.186971	POB, HDGREE, CIP2021, noc21, KOL, AGEIMM, Gender, MarStH, HHSIZE
noc21	5.575e+00	8	1.113375	POB, HDGREE, CIP2021, LOCSTUD, KOL, AGEIMM, Gender, MarStH, HHSIZE
KOL	1.358e+00	1	1.165291	POB, HDGREE, CIP2021, LOCSTUD, noc21, AGEIMM, Gender, MarStH, HHSIZE
AGEIMM	9.767e+00	12	1.099613	POB, HDGREE, CIP2021, LOCSTUD, noc21, KOL, Gender, MarStH, HHSIZE
Gender	1.375e+00	1	1.172563	POB, HDGREE, CIP2021, LOCSTUD, noc21, KOL, AGEIMM, MarStH, HHSIZE
MarStH	2.207e+00	5	1.082369	POB, HDGREE, CIP2021, LOCSTUD, noc21, KOL, AGEIMM, Gender, HHSIZE
HHSIZE	1.380e+00	1	1.174656	POB, HDGREE, CIP2021, LOCSTUD, noc21, KOL, AGEIMM, Gender, MarStH

$$\log(TotInc) = \beta_0 + POB\beta_1 + X_2\beta_2 + \cdots + X_{13}\beta_{13} + \epsilon$$

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Discussion

Table 5—: Summarized Table of Significant Results

Variable	Estimate	Robust SE
POBUK	4.591***	1.200
POBMiddle East + Western and Central Asia	4.297**	1.568
POBSoutheast Asia	-3.211**	1.237
POBPoland × HDGREEApprenticeship Certificate	6.006***	1.666
POBPhilippines × HDGREEApprenticeship Certificate	4.169**	1.366
POBVietnam × HDGREEApprenticeship Certificate	7.298***	1.141
POBUK × HDGREEBachelor's Degree	-4.448***	1.145
POBSoutheast Asia × HDGREEBachelor's Degree	3.944**	1.353
POBIndia × HDGREEBachelor's Degree	-4.105***	0.918
HDGREENon-apprenticeship Trades × AGEIMM20 to 24 yo	-3.221**	1.077
HDGREENon-apprenticeship Trades × AGEIMM35 to 39 yo	-2.380***	0.660
POBSoutheast Asia × LOCSTUDUS	-1.525**	0.545
POBNorthern + Western Europe × LOCSTUDEurope	2.860***	0.727
POBSouth + West Africa × LOCSTUDEurope	2.951***	0.765
POBEast Asia × LOCSTUDEurope	2.585**	0.899
POBPakistan × LOCSTUDEurope	3.860**	1.231
Number of Observations	1435	0.413
R^2 Adj.	0.100	1.09

The positive estimates of immigrants from Europe, the Middle East, and most parts of Asia indicate that they have a higher income compared to Canadians, while immigrants from Southeast Asia have lower total income levels.

Apprentice Certificate: immigrants from Southern Europe and Southeast Asia with an apprenticeship certificate tend to have a higher income than Canadians with an apprenticeship certificate by between 4% to 7%

Education: immigrants from Southeast Asia with a Bachelor's degree tend to earn more than Canadians with a Bachelor's by 3.9%. Indian and UK immigrants with the same degree earn less by 4.1% to 4.4%

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Age: immigrants who immigrated to Canada between the ages of 20 and 40 with an education in non-apprenticeship trades experienced between 2.3% to 3.2% less income than those who immigrated at a younger age with the same educational background.

Place of Education: immigrants from Southeast Asia and East Asia who studied in Europe tend to have a higher income than immigrants who studied in Canada by between 2.5% to 4%. This result contradicts the idea where foreign education is a disadvantage, especially for Southeast Asian and East Asian immigrants who may have language barriers.

Conclusion

- This paper determined that region of origin is a significant and persistent factor in explaining income disparities among immigrants in Canada.
- Immigrants from Western Europe and North America tend to earn more than those from regions such as Southeast Asia and Eastern Africa, even after controlling for factors such as: education, language proficiency, occupation, and age at immigration.
- There are strong associations between foreign education, age of arrival, and total income, with interaction terms suggesting that the income returns to education vary substantially by region and time if immigration.